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Medium processing apparatus, image forming system, medium processing method, and non-transitory recording medium

Abstract

A medium processing apparatus includes a conveyor, a receptacle, an aligner, a crimp binder, and circuitry. The conveyor conveys a medium. The receptacle holds the medium. The aligner contacts a downstream end of the medium and aligns a position of the medium in a conveyance direction of the medium. The crimp binder faces the downstream end of the medium to press and deform a binding position on media of which the position is aligned, to bind the media. The aligner and the crimp binder are movable in a main scanning direction orthogonal to each of the conveyance direction and a thickness direction of the medium. When determining that the aligner overlaps the binding position when the receptacle is viewed in the thickness direction, the circuitry moves the aligner in the main scanning direction away from the binding position and causes the crimp binder to crimp and bind the media.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application Nos. 2022-121881, filed on Jul. 29, 2022, and 2023-086434, filed on May 25, 2023, in the Japan Patent Office, the entire disclosure of each of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

(2) Embodiments of the present disclosure relate to a medium processing apparatus, an image forming system, a medium processing method, and a non-transitory recording medium.

Related Art

(3) Medium processing apparatuses are known in the related art that bind, into a bundle, sheet-shaped media on which images are formed by image forming apparatuses. Since sheets of paper are widely known as an example of sheet-shaped media, a “sheet bundle” that is a stack of sheets of paper is used as an example of a bundle of sheet-shaped media in the following description. Some medium processing apparatuses include a crimper that can perform so-called “crimp binding” without metal binding needles from a viewpoint of resource saving and reduction in environmental load. Specifically, the crimper sandwiches a sheet bundle with serrate binding teeth to press and deform the sheet bundle.

SUMMARY

(4) According to an embodiment of the present disclosure, a medium processing apparatus includes a conveyor, a receptacle, a conveyance-direction aligner, a crimp binder, and circuitry. The conveyor conveys a sheet-shaped medium in a conveyance direction. The receptacle holds the medium conveyed by the conveyor. The conveyance-direction aligner contacts a downstream end, in the conveyance direction, of the medium placed on the receptacle and aligns a position, in the conveyance direction, of the medium. The crimp binder is disposed to face the downstream end, in the conveyance direction, of the medium placed on the receptacle to press and deform a binding position on a plurality of media, including the medium, of which the position in the conveyance direction is aligned by the conveyance-direction aligner, to bind the plurality of media. The circuitry controls operations of the conveyor, the conveyance-direction aligner, and the crimp binder. The conveyance-direction aligner and the crimp binder are movable in a main scanning direction orthogonal to each of the conveyance direction and a thickness direction of the medium placed on the receptacle. The circuitry determines whether the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the circuitry determines that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction, the circuitry moves the conveyance-direction aligner in the main scanning direction to a position away from the binding position. The circuitry then causes the crimp binder to crimp and bind the plurality of media.

(5) According to an embodiment of the present disclosure, a novel image forming system includes an image forming apparatus and the medium processing apparatus. The image forming apparatus forms an image on a medium. The medium processing apparatus crimps and binds a plurality of media, including the medium, on each of which the image is formed by the image forming apparatus.

(6) According to an embodiment of the present disclosure, a novel medium processing method includes determining whether a conveyance-direction aligner overlaps a binding position when a receptacle is viewed in a thickness direction of a medium, moving the conveyance-direction aligner in a main scanning direction to a position away from the binding position based on a determination that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction of the medium, the main scanning direction being a direction orthogonal to each of a conveyance direction of the medium and the thickness direction of the medium placed on the receptacle, and moving a crimp binder in the main scanning direction and causing the crimp binder to crimp and bind a plurality of media including the medium.

(7) According to an embodiment of the present disclosure, a novel non-transitory recording medium stores a plurality of instructions which, when executed by one or more processors, causes the processors to perform the medium processing method.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete appreciation of embodiments of the present disclosure and many of the

attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

- (2) FIG. 1 is a diagram illustrating the overall configuration of an image forming system according to an embodiment of the present disclosure;
- (3) FIG. 2 is a diagram illustrating an internal configuration of a post-processing apparatus in the image forming system of FIG. 1;
- (4) FIG. 3 is a schematic view of an upstream side of a crimp binder of the post-processing apparatus of FIG. 2 in a conveyance direction;
- (5) FIGS. 4A and 4B are schematic diagrams illustrating a configuration of the crimp binder of FIG. 3;
- (6) FIG. 5 is a schematic view of a downstream upstream side of a stapling unit of the post-processing apparatus of FIG. 2 in a conveyance direction;
- (7) FIGS. 6A and 6B are views of an internal tray of the post-processing apparatus of FIG. 2 in a thickness direction of a sheet;
- (8) FIG. 7 is a block diagram illustrating a hardware configuration of the post-processing apparatus of FIG. 2 to control the post-processing apparatus;
- (9) FIG. 8 is a flowchart of a binding process performed by the crimp binder of FIG. 3;
- (10) FIGS. 9A to 9F are diagrams illustrating an example of the positions of end fences and the crimp binder of FIG. 3 during the binding process of FIG. 8;
- (11) FIGS. 10A to 10E are diagrams illustrating another example of the positions of end fences and the crimp binder of FIG. 3 during the binding process of FIG. 8;
- (12) FIGS. 11A to 11C are diagrams illustrating variations in intervals between binding positions in a main scanning direction;
- (13) FIG. 12 is a graph illustrating a relation between a binding strength and intervals between binding positions;
- (14) FIGS. 13A to 13C are diagrams illustrating the positions of side fences during a binding process according to a first modification;
- (15) FIGS. 14A and 14B are diagrams illustrating the positions of a tapping roller during a binding process according to a second modification;
- (16) FIGS. 15A to 15C are diagrams illustrating the positions of side fences during a binding process according to a third modification; and
- (17) FIGS. 16A to 16C are diagrams illustrating the positions of an internal tray and side fences during a binding process according to a fourth modification.
- (18) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

- (19) In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.
- (20) Referring now to the drawings, embodiments of the present disclosure are described below.
- (21) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
- (22) For the sake of simplicity, like reference numerals are given to identical or corresponding constituent elements such as parts and materials having the same functions, and redundant descriptions thereof are omitted unless otherwise required.
- (23) As used herein, the term “connected-coupled” includes both direct connections and

connections in which there are one or more intermediate connecting elements.

(24) With reference to the drawings, a description is now given of an image forming system **1** according to an embodiment of the present disclosure.

(25) FIG. **1** is a diagram illustrating the overall configuration of the image forming system **1**.

(26) The image forming system **1** has a function of forming an image on a sheet P as a sheet-shaped medium and performing post-processing on the sheet P on which the image is formed. As illustrated in FIG. **1**, the image forming system **1** includes an image forming apparatus **2** and a post-processing apparatus **3** serving as a medium processing apparatus.

(27) The image forming apparatus **2** forms an image on the sheet P and outputs the sheet P bearing the image to the post-processing apparatus **3**. The image forming apparatus **2** includes a tray that accommodates the sheet P, a conveyor that conveys the sheet P accommodated in the tray, and an image forming device that forms an image on the sheet P conveyed by the conveyor. The image forming device may be an inkjet image forming device that forms an image with ink or an electrophotographic image forming device that forms an lineage with toner. Since the image forming apparatus **2** has a typical configuration, a detailed description of the configuration and functions of the image forming apparatus **2** will be omitted unless otherwise required.

(28) FIG. **2** is a diagram illustrating an internal configuration of the post-processing apparatus **3**.

(29) The post-processing apparatus **3** performs post-processing on the sheet P on which an image is formed by the image forming apparatus **2**. The post-processing according to the present embodiment is binding as a process to bind the sheets P on each of which an image is formed as a bundle of sheets P. In the following description, the bundle of sheets P may be referred to as a “sheet bundle Pb” serving as a bundle of media. More specifically, the binding according to the present embodiment includes so-called “crimp binding” and “stapling.” The crimp binding is a process to press and deform the sheet bundle Pb to bind the sheet bundle Pb. The stapling is a process to pass a staple through the sheet bundle Pb to bind the sheet bundle Pb. The binding includes edge binding and saddle binding. The edge binding is a process to bind an end (including an edge) of the sheet bundle Pb. The saddle binding is a process to bind the center of the sheet bundle Pb.

(30) The post-processing apparatus **3** includes conveyance roller pairs **10** to **19** serving as conveyors and a switching claw **20**. The conveyance roller pairs **10** to **19** convey, inside the post-processing apparatus **3**, the sheet P supplied from the image forming apparatus **2**. More specifically, the conveyance roller pairs **10** to **13** convey the sheet P along a first conveyance passage Ph1. The conveyance roller pairs **14** and **15** convey the sheet P along a second conveyance passage Ph2. The conveyance roller pairs **16** to **19** convey the sheet P along a third conveyance passage Ph3.

(31) The first conveyance passage Ph1 is a passage extending to a first output tray **21** from a supply port through which the sheet P is supplied from the image forming apparatus **2**. The second conveyance passage Ph2 is a passage branching from the first conveyance passage Ph1 between the conveyance roller pairs **11** and **14** in a conveyance direction and extending to a second output tray **30** via an internal tray **22**. The third conveyance passage Ph3 is a passage branching from the first conveyance passage Ph1 between the conveyance roller pairs **11** and **14** in the conveyance direction and extending to a third output tray **36**.

(32) The switching claw **20** is disposed at a branching position of the first conveyance passage Ph1 and the second conveyance passage Ph2.

(33) The switching claw **20** can be switched between a first position and a second position. The switching claw **20** in the first position guides the sheet P to be output to the first output tray **21** through the first conveyance passage Ph1. The switching claw **20** in the second position guides the sheet P conveyed through the first conveyance passage Ph1 to the second conveyance passage Ph2. When a trailing end of the sheet P entering the second conveyance passage Ph2 passes through the conveyance roller pair **11**, the conveyance roller pair **14** is rotated in the reverse direction to guide the sheet P to the third conveyance passage Ph3. The post-processing apparatus **3** further includes a

plurality of sensors that detects the positions of the sheet P in the first conveyance passage Ph1, the second conveyance passage Ph2, and the third conveyance passage Ph3. Each of the plurality of sensors is indicated by a black triangle mark in FIG. 2.

(34) The post-processing apparatus 3 includes the first output tray 21. The sheet P output through the first conveyance passage Ph1 is placed on the first output tray 21. Among the sheets P supplied from the image forming apparatus 2, the sheets P that are not bound are output to the first output tray 21.

(35) The post-processing apparatus 3 further includes the internal tray 22 serving as a receptacle, a tapping roller 23 serving as a contact and separation device, a return roller 24, an end fence 25 serving as a conveyance-direction aligner, side fences 26L and 26R serving as width-direction aligners, a crimp binder 27, a stapling unit 28, a release claw 29 serving as a releaser, the second output tray 30, and a filler 31. The internal tray 22, the tapping roller 23, the return roller 24, the end fence 25, the side fences 26L and 26R, the crimp binder 27, the stapling unit 28, and the release claw 29 perform the edge binding on the sheet bundle Pb constructed of the sheets P conveyed through the second conveyance passage Ph2. Among the sheets P supplied from the image forming apparatus 2, the sheet bundle Pb subjected to the edge binding is output to the second output tray 30.

(36) The “edge binding” includes “parallel binding,” “oblique binding,” and “vertical binding.” The “parallel binding” is a process to bind the sheet bundle Pb along one side of the sheet bundle Pb parallel to a main scanning direction. The “oblique binding” is a process to bind a corner of the sheet bundle Pb. The “vertical binding” is a process to bind the sheet bundle Pb along one side of the sheet bundle Pb parallel to the conveyance direction.

(37) In the following description, a direction in which the sheet P is conveyed from the conveyance roller pair 15 toward the end fence 25 is defined as a “conveyance direction” of the sheet P. In other words, the “conveyance direction” corresponds to a direction in which the sheet P that has been output from the image forming apparatus 2 is moved toward the end fence 25 by the conveyance roller pair 15 after being moved toward the second output tray 30 by, for example, the conveyance roller pair 10. A direction that is orthogonal to each of the conveyance direction and a thickness direction of the sheet P is defined as a “main scanning direction” or a “width direction of the sheet P.”

(38) The internal tray 22 is disposed downstream from the conveyance roller pair 15 in the conveyance direction. The sheets P that are sequentially conveyed through the second conveyance passage Ph2 are temporarily placed on the internal tray 22 serving as a receptacle. The internal tray 22 according to the present embodiment is inclined downward in the conveyance direction. The tapping roller 23 is rotatably held at an end of a rotary arm above the internal tray 22. The tapping roller 23 comes into contact with and separates from the uppermost sheet P placed on the internal tray 22 by the rotation of the rotary arm. The return roller 24 is disposed above the internal tray 22, downstream from the tapping roller 23 in the conveyance direction.

(39) When the rotary arm is pivoted in a direction in which the tapping roller 23 is separated from the internal tray 22, the sheet P conveyed in the conveyance direction by the conveyance roller pair 15 enters the internal tray 22. In this state, when the rotary arm is pivoted in a direction in which the tapping roller 23 approaches the internal tray 22, the tapping roller 23 abuts, from above, against the sheet P conveyed onto the internal tray 22 by the conveyance roller pair 15. The sheet P in contact with the tapping roller 23 is separated from the conveyance roller pair 15 and placed on the internal tray 22. The return roller 24 contacts the upper face of the sheet P placed on the internal tray 22 and rotates to guide the sheet P toward the end fence 25.

(40) The end fence 25 is disposed downstream from the internal tray 22 in the conveyance direction. The end fence 25 is movable in the main scanning direction along the surface of the sheet P or the sheet bundle Pb placed on the internal tray 22. The end fence 25 contacts a downstream end, in the conveyance direction, of the sheet P or the sheet bundle Pb placed on the internal tray 22

to align the downstream end, in the conveyance direction, of the plurality of sheets P of the sheet bundle Pb. The side fences **26L** and **26R** are disposed on opposed sides on the internal tray **22** in the main scanning direction. The side fences **26L** and **26R** are movable in the main scanning direction. The side fences **26L** and **26R** contact opposed ends, in the main scanning direction, of the sheet P or the sheet bundle Pb placed on the internal tray **22** to align the ends, in the main scanning direction, of the plurality of sheets P of the sheet bundle Pb.

(41) The crimp binder **27** and the stapling unit **28** are disposed downstream from the internal tray **22** in the conveyance direction. The crimp binder **27** and the stapling unit **28** are independently movable in the main scanning direction along the surface of the sheet P the sheet bundle Pb placed on the internal tray **22**. The crimp binder **27** and the stapling unit **28** perform the edge binding on an end of the sheet bundle Pb aligned by the end fence **25** and the side fences **26L** and **26R**.

(42) More specifically, the crimp binder **27** sandwiches the binding position on the sheet bundle Pb placed on the internal tray **22** with senate binding teeth from both sides in the thickness direction. As a result, the binding position on the sheet bundle Pb is pressed and deformed. Thus, the sheet bundle Pb is crimped and bound.

(43) FIG. **3** is a schematic view of an upstream side of the crimp binder **27** in the conveyance direction.

(44) As illustrated in FIG. **3**, a guide shaft **37** extends in the main scanning direction at a position downstream from the internal tray **22** in the conveyance direction. The crimp binder **27** is moved in the main scanning direction along the surface of the sheet bundle Pb placed on the internal tray **22**, in other words, along the guide shaft **37**, by a driving force that is transmitted from a crimper movement motor **238** by a driving force transmission assembly **240** including pulleys **240a** and **240b** and a timing belt **240c**. A crimper shaft **340** provided with a drive transmission gear **340a** is fixed to a bottom face of a crimping frame **27c** that holds the components of the crimp binder **27** such as upper crimping teeth **27a** and lower crimping teeth **27b**. The crimper shaft **340** and the drive transmission gear **340a** are held by a base **48** on which the crimping frame **27c** is disposed, so as to be rotatable in the forward and reverse directions. The drive transmission gear **340a** meshes with an output gear **239a** of a crimper pivot motor **239**. The crimp binder **27** is rotated in the forward and reverse directions on the base **48** about the crimper shaft **340** extending in the thickness direction of the sheet P placed on the internal tray **22**, by a driving force transmitted from the crimper pivot motor **239** to the crimper shaft **340** via the output gear **239a** and the drive transmission gear **340a**. The guide shaft **37**, the crimper movement motor **238**, the crimper pivot motor **239**, the crimper shaft **340**, and the driving force transmission assembly **240** construct a driving assembly of the crimp binder **27**.

(45) As illustrated in FIGS. **4A** and **4B**, the crimp binder **27** includes a pair of binding teeth (i.e., the upper crimping teeth **27a** and the lower crimping teeth **27b**). The upper crimping teeth **27a** and the lower crimping teeth **27b** are disposed to face each other in the thickness direction of the sheet bundle Pb so as to sandwich the sheet bundle Pb placed on the internal tray **22**. The upper crimping teeth **27a** and the lower crimping teeth **27b** have respective serrate faces facing each other. The senate face of each of the upper crimping teeth **27a** and the lower crimping teeth **27b** includes concave portions and convex portions alternately formed. The concave portions and the convex portions of the upper crimping teeth **27a** are shifted from those of the lower crimping teeth **27b** such that the upper crimping teeth **27a** are engaged with the lower crimping teeth **27b**. The upper crimping teeth **27a** and the lower crimping teeth **27b** are brought into contact with and separated from each other by a driving force of a contact-separation motor **27d** illustrated in FIG. **7**.

(46) In the process of supplying the sheets P of the sheet bundle Pb to the internal tray **22**, the upper crimping teeth **27a** and the lower crimping teeth **27b** are apart from each other as illustrated in FIG. **4A**. When all the sheets P of the sheet bundle Pb are placed on the internal tray **22**, the upper crimping teeth **27a** and the lower crimping teeth **27b** are engaged with each other to press and deform the sheet bundle Pb in the thickness direction as illustrated in FIG. **4B**. As a result, the sheet

bundle Pb that has been placed on the internal tray 22 is crimped and bound. The sheet bundle Pb thus crimped and bound is output to the second output tray by the conveyance roller pair 15.

(47) The configuration of the crimp binder 27 as a crimping assembly is not limited to the configuration of a moving assembly exemplified in the present embodiment, provided that the upper crimping teeth 27a and the lower crimping teeth 27b of the crimping assembly are engaged with each other. For example, the crimping assembly may be a crimping assembly disclosed in Japanese Patent No. 6057167 or its corresponding U.S. Patent Application Publication No. 2014-0219747, which is hereby incorporated by reference as though disclosed herein in its entirety. In this case, the crimping assembly brings the upper crimping teeth 27a and the lower crimping teeth 27b into contact with each other and separates the upper crimping teeth 27a and the lower crimping teeth 27b from each other with a link assembly and a driving source that simply rotates forward or that rotates forward and backward. Alternatively, the crimping assembly may employ a linear motion system to linearly bring the upper crimping teeth 27a and the lower crimping teeth 27b into contact with each other and separate the upper crimping teeth 27a and the lower crimping teeth 27b from each other with a screw assembly that converts the forward and backward rotational motions of a driving source into linear reciprocating motion.

(48) Now, a detailed description is given of the stapling unit 28.

(49) The stapling unit 28 executes stapling or a stapling process. Specifically, the stapling unit 28 passes a staple through a binding position on the sheet bundle Pb placed on the internal tray 22 to bind the sheet bundle Pb.

(50) FIG. 5 is a schematic view of a downstream side of the stapling unit 28 in the conveyance direction.

(51) The stapling unit 28 includes a stapler 62 that binds the sheet bundle Pb with a staple or staples. The stapler 62 is disposed downstream from the internal tray 22 in the conveyance direction and apart from the crimp binder 27 in the main scanning direction.

(52) The stapler 62 has a configuration for performing so-called “stapling” (i.e., stapling process) to bind the sheet bundle Pb with a staple or staples. More specifically, the stapler 62 includes a stapling-part drive motor 62d illustrated in FIG. 7. The stapling-part drive motor 62d drives a stapling part 62a. A driving force of the stapling-part drive motor 62d causes a staple loaded in the stapling part 62a to pass through the sheet bundle Pb. Thus, the stapling part 62a binds the sheet bundle Pb. Since the stapler 62 has a typical configuration, a detailed description thereof will be omitted unless otherwise required.

(53) As illustrated in FIG. 5, the stapling unit 28 includes a stapling-unit movement assembly 77.

(54) The stapling-unit movement assembly 77 moves the stapling unit 28 in the main scanning direction along the downstream end, in the conveyance direction, of the sheet P or the sheet bundle Pb placed on the internal tray 22.

(55) The stapling-unit movement assembly 77 includes, for example, a base 78, the guide shaft 37, a stapling-unit movement motor 80, and a driving force transmission assembly 81. The driving force transmission assembly 81 transmits a driving force of the stapling-unit movement motor 80 to the base 78 via pulleys 81a and 81b and a timing belt 81c. A stapler shaft 83 provided with a drive transmission gear 83a is fixed to a bottom face of a stapling frame 62b that holds the components of the stapler 62. The stapler shaft 83 and the drive transmission gear 83a are held by the base 78 on which the stapling frame 62b is disposed, so as to be rotatable in the forward and reverse directions. The drive transmission gear 83a meshes with an output gear 82a of a stapler pivot motor 82. The stapler 62 can be rotated in the forward and reverse directions about the stapler shaft 83 on the base 78 by a driving force transmitted from the stapler pivot motor 82 to the stapler shaft 83 via the output gear 82a and the drive transmission gear 83a.

(56) The crimp binder 27 and the stapling unit 28 are supported by the common guide shaft 37. The driving force transmission assembly 240 and the stapling-unit movement assembly 77 move the crimp binder 27 and the stapling unit 28 in the main scanning direction along the common guide

shaft 37. The driving force transmission assembly 240 and the stapling-unit movement assembly 77 can independently move the crimp binder 27 and the stapling unit 28.

(57) The release claw 29 is disposed downstream from the internal tray 22 in the conveyance direction. The position, in the main scanning direction, of the release claw 29 is fixed. The release claw 29 is movable upstream in the conveyance direction (i.e., toward the conveyance roller pair 15) in contact with the downstream end, in the conveyance direction, of the sheet bundle Pb placed on the internal tray 22. The release claw 29 moving upstream in the conveyance direction moves the sheet bundle Pb subjected to the edge binding upstream in the conveyance direction along the internal tray 22. As a result, the sheet bundle Pb subjected to the edge binding is released from the internal tray 22 and enters between rollers of the conveyance roller pair 15. Then, the conveyance roller pair 15 outputs the sheet bundle Pb subjected to the edge binding to the second output tray 30.

(58) The second output tray 30 is disposed on an outer side face of a housing of the post-processing apparatus 3 such that the second output tray 30 can vertically move.

(59) The filler 31 is rotatable above the second output tray 30. An end of the filler 31 contacts the sheet bundle Pb placed on the second output tray 30. The filler 31 detects that the height (thickness) of the sheet bundle Pb stacked on the second output tray 30 reaches a threshold. The filler 31 then outputs the detection result to a controller 100 illustrated in FIG. 7. A detailed description of the controller 100 is deferred. When the filler 31 detects that the height of the sheet bundle Pb reaches the threshold, the controller 100 lowers the second output tray 30 by a given amount.

(60) The post-processing apparatus 3 further includes an end fence 32, a saddle binder 33, a sheet folding blade 34, a hole punch 35, and the third output tray 36. The end fence 32, the saddle binder 33, and the sheet folding blade 34 perform the saddle binding on the sheet bundle Pb constructed of the sheets P that are conveyed through the third conveyance passage Ph3. Among the sheets P supplied from the image forming apparatus 2, the sheet bundle Pb subjected to the saddle binding is output to the third output tray 36.

(61) The end fence 32 aligns the positions, in a direction in which the sheets P are conveyed, of the sheets P that are sequentially conveyed through the third conveyance passage Ph3. The end fence 32 is movable between a binding position where the end fence 32 causes the center of the sheet bundle Pb to face the saddle binder 33 and a folding position where the end fence 32 causes the center of the sheet bundle Pb to face the sheet folding blade 34. The saddle binder 33 binds the center of the sheet bundle Pb aligned by the end fence 32 at the binding position. The sheet folding blade 34 folds, in half, the sheet bundle Pb placed on the end fence 32 at the folding position and causes the conveyance roller pair 18 to sandwich the sheet bundle Pb. The conveyance roller pairs 18 and 19 output the sheet bundle Pb subjected to the saddle binding to the third output tray 36. The hole punch 35 punches a through hole in the sheet bundle Pb that is conveyed by the conveyance roller pairs 15 and 19.

(62) Each of FIGS. 6A and 6B is a view of the internal tray 22 in the thickness direction of the sheet P.

(63) As illustrated in FIGS. 6A and 6B, the post-processing apparatus 3 according to the present embodiment includes a symmetrical pair of end fences 25L and 25R. The end fences 25L and 25R are arranged opposite each other across the release claw 29 in the main scanning direction. In other words, the end fences 25L and 25R are arranged opposite each other across a center position C, in the main scanning direction, of the sheet bundle Pb placed on the internal tray 22.

(64) The end fences 25L and 25R are disposed at equal distances from the release claw 29. As described above, the crimp binder 27 and the stapling unit 28 are supported by the guide shaft 37 extending in the main scanning direction at a position downstream from the internal tray 22 in the conveyance direction, such that the crimp binder 27 and the stapling unit 28 can independently move in the main scanning direction. In FIGS. 6A and 6B and subsequent figures, illustration of the stapling unit 28 may be omitted.

(65) The end fences **25L** and **25R** according to the present embodiment move between respective proximate positions illustrated in FIGS. **9A** to **9C** and respective distant positions illustrated in FIGS. **9D** to **9F**. The interval between the end fences **25L** and **25R** in the main scanning direction is greater at the respective distant positions than at the respective proximate positions. In other words, the distant position is farther from the center position **C** than the proximate position. Note that the distance from the center position **C** to the end fence **25L** at the proximate position is equal to the distance from the center position **C** to the end fence **25R** at the proximate position. Similarly, the distance from the center position **C** to the end fence **25L** at the distant position is equal to the distance from the center position **C** to the end fence **25R** at the distant position. Thus, the end fences **25L** and **25R** move in the main scanning direction while maintaining the same distance from the center position **C**. In other words, the interval between the end fences **25L** and **25R** in the main scanning direction increases or decreases. The positions to which the end fences **25L** and **25R** can move are not limited to the positions illustrated in FIGS. **9A** to **9F**.

(66) On the other hand, the crimp binder **27** according to the present embodiment moves between a standby position **HP** illustrated in FIG. **9A** and positions where the crimp binder **27** faces binding positions **S1** to **S4** as illustrated in FIGS. **9B** to **9F**. The standby position **HP** is away in one of the main scanning directions from the sheet **P** placed on the internal tray **22**. For example, in FIG. **9A**, the standby position **HP** is distanced to the right of the sheet **P** in the main scanning direction. The binding positions **S1** to **S4** are positions on the sheet bundle **Pb** placed on the internal tray **22**.

However, the specific positions of the binding positions **S1** to **S4** are not limited to the positions illustrated in FIGS. **9A** and **9F**. The binding positions **S1** to **S4** may be positions in the main scanning direction at the downstream end, in the conveyance direction, of the sheet bundle **Pb**.

(67) As an example, as illustrated in FIG. **6A**, the end fences **25L** and **25R** are independently moved in the main scanning direction by a driving force transmitted from end-fence motors **38L** and **38R** serving as first driving sources through a driving force transmission assemblies **39L** and **39R**. The driving force transmission assembly **39L** includes, for example, pulleys **39La** and a timing belt **39Lb**. Similarly, the driving force transmission assembly **39R** includes, for example, pulleys **39Ra** and a timing belt **39Rb**. The position, in the main scanning direction, of the end fence **25L** is detected by a sensor disposed within the moving range of the end fence **25L**, a rotary encoder **38Le** (see FIG. **7**) of the end-fence motor **38L**, or a combination thereof. Similarly, the position of the end fence **25R** in the main scanning direction is detected by a sensor disposed within the moving range of the end fence **25R**, a rotary encoder **38Re** (see FIG. **7**) of the end-fence motor **38R**, or a combination thereof. The readings indicating the detected positions of the end fences **25L** and **25R** are provided to the controller **100**.

(68) As described above, the crimp binder **27** is moved in the main scanning direction by the driving force that is transmitted from the crimper movement motor **238** by the driving force transmission assembly **240** including the pulleys **240a** and **240b** and the timing belt **240c**. On the other hand, the stapling unit **28** is moved in the main scanning direction by the driving force that is transmitted from the stapling-unit movement motor **80** by the driving force transmission assembly **81** including the pulleys **81a** and **81b** and the timing belt **81c**. The positions, in the main scanning direction, of the crimp binder **27** and the stapling unit **28** are detected by a rotary encoder **238e** (see FIG. **7**) of the crimper movement motor **238** and a rotary encoder **80e** (see FIG. **7**) of the stapling-unit movement motor **80**, respectively. Readings indicating the detected positions of the crimp binder **27** and the stapling unit **28** are provided to the controller **100**.

(69) As another example, as illustrated in FIG. **6B**, the driving force transmission assemblies **39L** and **39R** and the driving force transmission assembly **240** may be coupled to each other. The end fences **25L** and **25R** and the crimp binder **27** may be moved in the main scanning direction in conjunction with each other by, for example, the driving force transmitted from the crimper movement motor **238** as a common driving source through the driving force transmission assemblies **39L** and **39R** and the driving force transmission assembly **240**. In this case, the end-

fence motors **38L** and **38R** illustrated in FIG. **6A** may be omitted.

(70) FIG. **7** is a block diagram illustrating a hardware configuration of the post-processing apparatus **3**.

(71) As illustrated in FIG. **7**, the post-processing apparatus **3** includes a central processing unit (CPU) **101**, a random access memory (RAM) **102**, a read only memory (ROM) **103**, a hard disk drive (HDD) **104**, and an interface (I/F) **105**. The CPU **101**, the RAM **102**, the ROM **103**, the HDD **104**, and the I/F **105** are connected to each other via a common bus **109**. The CPU **101** is an arithmetic unit and controls the overall operation of the post-processing apparatus **3**. The RAM **102** is a volatile storage medium that allows data to be read and written at high speed. The CPU **101** uses the RAM **102** as a work area for data processing. The ROM **103** is a read-only non-volatile storage medium that stores programs such as firmware. The HDD **104** is a non-volatile storage medium that allows data to be read and written and has a relatively large storage capacity. The HDD **104** stores, for example, an operating system (OS), various control programs, and application programs.

(72) By an arithmetic function of the CPU **101**, the post-processing apparatus **3** processes, for example, a control program stored in the ROM **103** and an information processing program (application program) loaded into the RAM **102** from a storage medium such as the HDD **104**. Such processing configures a software controller including various functional modules of the post-processing apparatus **3**. The software controller that is thus configured cooperates with hardware resources of the post-processing apparatus **3** to construct functional blocks that implement functions of the post-processing apparatus **3**. In other words, the CPU **101**, the RAM **102**, the ROM **103**, and the HDD **104** construct the controller **100** that controls the operation of the post-processing apparatus **3**.

(73) The I/F **105** is an interface that connects the conveyance roller pairs **10**, **11**, **14**, and the switching claw **20**, the tapping roller **23**, the side fences **26L** and **26R**, the end-fence motors **38L** and **38R**, the contact-separation motor **27d**, the crimper pivot motor **239**, the crimper movement motor **238**, the stapling-part drive motor **62d**, the stapler pivot motor **82**, the stapling-unit movement motor **80**, the release claw **29**, the rotary encoders **38Le**, **38Re**, **238e**, and **80e**, the filler **31**, and a control panel **110** to the common bus **109**.

(74) The controller **100** controls, via the I/F **105**, the operations of the conveyance roller pairs **10**, **11**, **14**, and **15**, the switching claw **20**, the tapping roller **23**, the side fences **26L** and **26R**, the end-fence motors **38L** and **38R**, the contact-separation motor **27d**, the crimper pivot motor **239**, the crimper movement motor **238**, the stapling-part drive motor **62d**, the stapler pivot motor **82**, the stapling-unit movement motor **80**, and the release claw **29**. On the other hand, the controller **100** acquires detection results from the rotary encoders **38Le**, **38Re**, **238e**, and **80e** and the filler **31**. Although FIG. **7** illustrates the components related to the stapling unit **28** and the crimp binder **27** that executes the edge binding, the components related to the saddle binder **33** that executes the saddle binding are controlled by the controller **100** like the components related to the stapling unit **28** and the crimp binder **27** that executes the edge binding.

(75) As illustrated in FIG. **1**, the image forming apparatus **2** includes the control panel **110**. The control panel **110** includes an operation unit that receives instructions from a user and a display serving as a notifier that notifies the user of information. The operation unit includes, for example, hard keys and a touch panel superimposed on the display. The control panel **110** acquires information from the operator through the operation unit and provides information to the operator through the display. Note that a specific example of the notifier is not limited to the display and may be a light emitting diode (LED) lamp or a speaker. The post-processing apparatus **3** may include the control panel **110** like the control panel **110** described above.

(76) FIG. **8** is a flowchart of a binding process.

(77) FIGS. **9A** to **9F** are diagrams illustrating an example of the positions of the end fences **25L** and **25R** and the crimp binder **27** during the binding process.

(78) For example, the controller **100** starts the binding process illustrated in FIG. **8** when the controller **100** acquires an instruction to execute the binding process from the image forming apparatus **2**. In the following description, the instruction to execute the binding process may be referred to as a “binding command.” The binding command includes, for example, the binding position on the sheet bundle Pb in the main scanning direction, the number of binding positions on the sheet bundle Pb, and the number of sheets P of the sheet bundle Pb. In the following description, the number of sheets P of the sheet bundle Pb may be referred to as the “number of sheets to be bound.” As illustrated in FIG. **9A**, at the start of the binding process, the end fences **25L** and **25R** are at the respective proximate positions while and the crimp binder **27** is at the standby position HP.

(79) In step **S501**, the controller **100** rotates the conveyance roller pairs **10**, **11**, **14**, and **15** to accommodate, in the internal tray **22**, the sheet P on which an image is formed by the image forming apparatus **2**. The controller **100** also moves the side fences **26L** and **26R** to align the position, in the main scanning direction, of the sheet P or the sheet bundle Pb placed on the internal tray **22**. In short, the controller **100** performs so-called jogging.

(80) Subsequently, in step **S502**, the controller **100** determines whether the number of sheets P accommodated in the internal tray **22** has reached the number of sheets to be bound, which is instructed by the binding command.

(81) When the controller **100** determines that the number of sheets P accommodated in the internal tray **22** has not reached the number of sheets to be bound (NO in step **S502**), the controller **100** executes the operation of step **S501** again. On the other hand, when the controller **100** determines that the number of sheets P accommodated in the internal tray **22** has reached the number of sheets to be bound (YES in step **S502**), in step **S503**, the controller **100** initializes a variable N stored in the HDD **104**. In other words, in step **S503**, the controller **100** sets the variable N to 1.

(82) Subsequently, in step **S504**, the controller **100** determines whether the end fence **25L** or **25R** overlaps a binding position N when the internal tray **22** is viewed in the thickness direction. In the present embodiment, as illustrated in FIGS. **9A** to **9F**, the binding positions **S1**, **S2**, **S3**, and **S4** are crimped and bound. The order in which the binding positions **S1** to **S4** are crimped and bound is not particularly limited. However, the binding position that overlaps neither the end fence **25L** nor **25R** is preferably crimped and bound first.

(83) When the controller **100** determines that neither the end fence **25L** nor **25R** overlaps the binding position **S1** (NO in step **S504**), in step **S506**, the controller **100** causes the crimp binder **27** to crimp and bind the binding position **S1** without executing the operation of step **S505** (i.e., without moving the end fence **25L** or **25R**). In other words, as illustrated in FIG. **9B** the controller **100** drives the crimper movement motor **238** to move the crimp binder **27** to the position where the crimp binder **27** faces the binding position **S1** and causes the crimp binder **27** to sandwich the binding position **S1** on the sheet bundle Pb between the upper crimping teeth **27a** and the lower crimping teeth **27b** to press and deform the binding position **S1**.

(84) Subsequently, in step **S507** the controller **100** determines whether any binding position that is yet to be crimped and bound exists.

(85) When the controller **100** determines that a binding position that is yet to be crimped and bound exists (YES in step **S507**), in step **S508**, the controller **100** increases the variable N by 1. Then, the controller **100** performs the operation of step **S504** again.

(86) Specifically, in step **S504**, the controller **100** determines whether the end fence **25L** or **25R** overlaps the binding position **S2**.

(87) When the controller **100** determines that neither the end fence **25L** nor **25R** overlaps the binding position **S2** (NO in step **S504**), in step **S506**, the controller **100** causes the crimp binder **27** to crimp and bind the binding position **S2** that overlaps neither the end fence **25L** nor **25R** as illustrated in FIG. **9C**.

(88) Thereafter, in step **S507**, the controller **100** determines again whether any binding position that

is yet to be crimped and bound exists.

(89) When the controller **100** determines that a binding position that is yet to be crimped and bound exists (YES in step S507), in step S508, the controller **100** increases the variable N by 1. Then, the controller **100** performs the operation of step S504 again.

(90) Specifically, in step S504, the controller **100** determines whether the end fence 25L or 25R overlaps the binding position S3.

(91) When the controller **100** determines that the end fence 25L or 25R overlaps the binding position S3 (YES in step S504), in step S505, the controller **100** drives the end-fence motors 38L and 38R to move the end fences 25L and 25R to the respective distant positions as illustrated in FIG. 9D. In other words, in step S505, the controller **100** moves the end fences 25L and 25R in the main scanning direction to respective positions away from the binding position S3.

(92) After moving the end fences 25L and 25R, in step S506, the controller **100** causes the crimp binder 27 to crimp and bind the binding position S3 as illustrated in FIG. 9E.

(93) Thereafter, in step S507, the controller **100** determines again whether any binding position that is yet to be crimped and bound exists.

(94) When the controller **100** determines that a binding position that is yet to be crimped and bound exists (YES in step S507), in step S508, the controller **100** increases the variable N by 1. Then, the controller **100** performs the operation of step S504 again.

(95) Specifically, in step S504, the controller **100** determines whether the end fence 25L or 25R overlaps the binding position S4.

(96) When the controller **100** determines that neither the end fence 25L nor 25R overlaps the binding position S4 (NO in step S504), in step S506, the controller **100** causes the crimp binder 27 to crimp and bind the binding position S4 that overlaps neither the end fence 25L nor 25R as illustrated in FIG. 9F.

(97) When the controller **100** determines that all the binding positions S1 to S4 are crimped and bound (NO in step S507), in step S509, the controller **100** outputs the sheet bundle Pb thus crimped and bound to the second output tray 30. More specifically, the controller **100** moves the release claw 29 upstream in the conveyance direction so that the rollers of the conveyance roller pair 15 sandwich the sheet bundle Pb. The controller **100** then rotates the conveyance roller pair 15 to output the sheet bundle Pb to the second output tray 30. The controller **100** also moves the end fences 25L and 25R and the crimp binder 27 to the respective initial positions illustrated in FIG. 9A.

(98) The number of binding positions is not limited to the number of binding positions illustrated in FIGS. 9A to 9F. The respective initial positions of the end fences 25L and 25R are not limited to the respective proximate positions.

(99) FIGS. 10A to 10E are diagrams illustrating another example of the positions of the end fences 25L and 25R and the crimp binder 27 during the binding process.

(100) In FIGS. 10A to 10E, six binding positions S1 to S6 are crimped and bound. The end fences 25L and 25R are at the respective distant positions at the start of the binding process.

(101) First, as illustrated in FIGS. 10A and 10B, the controller **100** causes the crimp binder 27 to sequentially crimp and bind the binding positions S1 to S4 that overlap neither the end fence 25L nor 25R. The controller **100** then moves the end fences 25L and 25R to the respective proximate positions away from the binding positions S5 and S6 as illustrated in FIG. 10C. The controller **100** then causes the crimp binder 27 to sequentially crimp and bind the binding positions S5 and S6 that overlap neither the end fence 25L nor 25R as illustrated in FIGS. 10D and 10E.

(102) Preferably, the binding positions are symmetrical with respect to the center position C as illustrated in FIGS. 9A to 10E. The controller **100** retracts the end fences 25L and 25R preferably after causing the crimp binder 27 to crimp and bind at least one binding position on each side of the center position C in the main scanning direction among the plurality of binding positions. The intervals between the plurality of binding positions in the main scanning direction is preferably

changed depending on, for example, the number of binding positions or the size of the sheet P in the main scanning direction.

(103) FIGS. **11A** to **11C** are diagrams illustrating variations in the intervals between the plurality of binding positions in the main scanning direction.

(104) FIG. **12** is a graph illustrating a relation between the binding strength and the intervals between the binding positions.

(105) As illustrated in FIG. **11A**, when the intervals are 0 mm between the adjacent binding positions in the main scanning direction, the binding strength is lower than a preferred binding strength as indicated by “none” in the axis representing the binding intervals in FIG. **12**. In other words, the plurality of binding positions are preferably arranged at given intervals in the main scanning direction.

(106) Even when the sheets P in the same size are conveyed, the widths of the sheets P in the main scanning direction vary between a case where the sheet P is conveyed with a longitudinal direction of the sheet P in the main scanning direction (i.e., long edge feed [LEF] illustrated in FIG. **11B**) and a case where the sheet P is conveyed with a short direction of the sheet P in the main scanning direction (i.e., short edge feed [SEF] illustrated in FIG. **11C**). When the sheets P in different sizes (for example, A4 and B5) are conveyed, the widths of the sheets P in the main scanning direction vary. To handle such a situation, the controller **100** may change the intervals between the plurality of binding positions in the main scanning direction depending on the width of the sheet P in the main scanning direction.

(107) Now, a description is given of two example ways to change the arrangement of the binding positions from the arrangement illustrated in FIG. **11B** to the arrangement illustrated in FIG. **11C**.

(108) As a first example, the intervals between the binding positions are narrowed with the binding positions closest to the center position C fixed. As a second example, the binding positions are shifted to the center position C while maintaining the intervals between the binding positions.

(109) As illustrated in FIG. **12**, when the intervals between the plurality of binding positions in the main scanning direction are equal to or greater than 5 mm as indicated by “large” in the axis representing the binding intervals in FIG. **12**, the binding strength decreases compared to a case where the intervals are less than 5 mm as indicated by “small” in the axis representing the binding intervals in FIG. **12**. For this reason, the intervals between the plurality of binding positions in the main scanning direction are preferably greater than 0 mm and smaller than 5 mm.

(110) Now, a description is given of some or all of the advantages according to the embodiment described above, the enumeration of which is not exhaustive or limiting.

(111) According to the embodiment described above, when the end fences **25L** and **25R**, which are movable in the main scanning direction, overlap the binding positions, the crimp binder **27** crimps and binds the binding positions after the end fences **25L** and **25R** are retracted. Thus, the crimp binder **27** crimps and binds selected positions in the main scanning direction on the sheet bundle Pb. Since a preferred number of binding positions secured, a preferred binding strength is maintained for the sheet bundle Pb.

(112) According to the embodiment described above, the binding strength is increased as illustrated in FIG. **12** by the crimp binding at a plurality of binding positions at given intervals in the main scanning direction. Thus, as compared with a case where the intervals between the binding positions are 0, a preferred binding strength is obtained for the sheet bundle Pb even when the number of binding positions is reduced.

(113) According to the embodiment described above, by changing the intervals between the binding positions in the main scanning direction depending on the width of the sheet P in the main scanning direction, a preferred number of binding positions is determined for the sheet bundles Pb in various sizes. As a result, a preferred binding strength is obtained for the sheet bundle Pb regardless of the size of the sheet bundle Pb.

(114) According to the embodiment described above, before the end fences **25L** and **25R** are moved

to positions where the end fences **25L** and **25R** do not overlap the binding positions, the crimp binder **27** crimps and binds at least one binding position, to prevent the misalignment of the sheet brindle **Pb** that may be caused by the movement of the end fences **25L** and **25R**.

(115) In the case where the end fences **25L** and **25R** and the crimp binder **27** are respectively driven by the end-fence motors **38L** and **38R** and the crimper movement motor **238** that are separate from each other as illustrated in FIG. **6A**, the positions and timings of movement of the end fences **25L** and **25R** and the crimp binder **27** are individually optimized. On the other hand, in the case where the end fences **25L** and **25R** and the crimp binder **27** are driven by the common crimper movement motor **238** as illustrated in FIG. **6B**, cost reduction and space saving of the post-processing apparatus **3** are achieved.

(116) Now, a description is given of a first modification of the embodiment described above.

(117) Specifically, referring now to FIGS. **13A** to **13C**, a description is given of an operation of the post-processing apparatus **3** according to the first modification.

(118) FIGS. **13A** to **13C** are diagrams illustrating the positions of the side fences **26L** and **26R** during a binding process according to the first modification.

(119) Note that detailed descriptions will be omitted of common features of the embodiment described above and the present modification. The following description concentrates on the differences between the embodiment described above and the present modification.

(120) According to the first modification, the controller **100** moves the side fences **26L** and **26R** so that the interval between the side fences **26L** and **26R** is greater than the width of the Sheet **P** in the main scanning direction as illustrated in FIG. **13A** when the sheet **P** is accommodated in the internal tray **22**. Thus, the sheet **P** smoothly enters the internal tray **22**. On the other hand, according to the first modification, the controller **100** moves the side fences **26L** and **26R** so that the interval between the side fences **26L** and **26R** is smaller than the width of the sheet **P** in the main scanning direction as illustrated in FIG. **13B**, before retracting the end fences **25L** and **25R** in step **S505** of FIG. **8** (in other words, before moving the end fences **25L** and **25R** in the main scanning direction). In other words, according to the first modification, the controller **100** retracts the end fences **25L** and **25R** in step **S505** of FIG. **8** while the interval between the side fences **26L** and **26R** is smaller than the width of the sheet **P** in the main scanning direction as illustrated in FIG. **13C**.

(121) According to the first modification, the controller **100** moves the end fences **25L** and in the main scanning direction while the sheet bundle **Pb** is fixed to the internal tray **22** by the side fences **26L** and **26R**, to prevent the position of the sheet bundle **Pb** from being shifted due to the movement of the end fences **25L** and **25R**.

(122) Since a desired position is crimped and bound by the crimp binder **27**, a preferred binding strength is obtained.

(123) Now, a description is given of a second modification of the embodiment described above.

(124) Specifically, referring now to FIGS. **14A** and **14B**, a description is given of an operation of the post-processing apparatus **3** according to the second modification. FIGS. **14A** and **14B** are diagrams illustrating the positions of the tapping roller **23** during a binding process according to the second modification.

(125) Note that detailed descriptions will be omitted of common features of the embodiment described above and the present modification. The following description concentrates on the differences between the embodiment described above and the present modification.

(126) According to the second modification, the controller **100** retracts the tapping roller **23** above the internal tray **22** as illustrated in FIG. **14A** when the sheet **P** sandwiched between the rollers of the conveyance roller pair **15** enters the internal tray **22**. According to the second modification, the controller **100** brings the tapping roller **23** into contact with the upper face of the sheet **P** that has entered the internal tray **22**, to separate the sheet **P** from the conveyance roller pair **15**. Thus, the sheet **P** conveyed by the conveyance roller pair **15** is smoothly accommodated in the internal tray

22.

(127) According to the second modification, the controller **100** brings the tapping roller **23** into contact with the uppermost sheet P placed on the internal tray **22** (in other words, the uppermost sheet P of the sheet bundle Pb) as illustrated in FIG. **14B**, before retracting the end fences **25L** and **25R** in step **S505** of FIG. **8**. In other words, according to the second modification, the controller **100** retracts the end fences **25L** and **25R** in step **S505** of FIG. **8** while the tapping roller **23** is in contact with the uppermost sheet P placed on the internal tray **22**.

(128) According to the second modification, the controller **100** moves the end fences **25L** and **25R** in the main scanning direction while the sheet bundle Pb is fixed to the internal tray **22** by the tapping roller **23**, to prevent the position of the sheet bundle Pb from being shifted due to the movement of the end fences **25L** and **25R**. Since a desired position is crimped and bound by the crimp binder **27**, a preferred binding strength is obtained.

(129) Now, a description is given of third and fourth modifications of the embodiment described above.

(130) Specifically, referring now to FIGS. **15A** to **16C**, a description is given of operations of the post-processing apparatus **3** according to the third and fourth modifications. FIGS. **15A** to **15C** are diagrams illustrating the positions of the side fences **26L** and **26R** during a binding process according to the third modification.

(131) FIGS. **16A** to **16C** are diagrams illustrating the positions of the internal tray **22** and the side fences **26L** and **26R** during a binding process according to the fourth modification.

(132) Note that detailed descriptions will be omitted of common features of the embodiment described above and the present modifications. The following description concentrates on the differences between the embodiment described above and the present modifications.

(133) According to the third and fourth modifications, in step **S504** of FIG. **8**, the controller **100** determines whether the end fence **25L** or **25R** overlaps the binding position N when the internal tray **22** is viewed in the thickness direction and whether the release claw **29** overlaps the binding position N when the internal tray **22** is viewed in the thickness direction. According to the third and fourth modifications, when the controller **100** determines that the release claw **29** overlaps the binding position **S3** as illustrated in FIGS. **15A** and **16A**, the controller **100** moves the sheet bundle Pb in the main scanning direction to a position where the release claw **29** is away from the binding position **S3** as illustrated in FIGS. **15B** and **16B**, before retracting the end fences **25L** and **25R** in step **S505** of FIG. **8**. In other words, according to the third and fourth modifications, the controller **100** causes the crimp binder **27** to crimp and bind the sheet bundle Pb in step **S506** of FIG. **8** after relatively moving the release claw **29** and the sheet bundle Pb in the main scanning direction so that the release claw **29** is away from the binding position **S3**, as illustrated in FIGS. **15C** and **16C**.

(134) According to the third modification illustrated in FIGS. **15A** to **15C**, the controller **100** moves the side fences **26L** and **26R** in one of the main scanning directions (rightward in the example of FIGS. **15A** to **15C**) relative to the internal tray **22**. As a result, the sheet bundle Pb is moved in the main scanning direction on the internal tray **22**. The relative movement of the release claw **29** to a position away from the binding position **S3** allows the crimp binder **27** to crimp and bind the sheet bundle Pb.

(135) On the other hand, according to the fourth modification illustrated in FIGS. **16A** to **16C**, the controller **100** moves the internal tray **22** and the side fences **26L** and **26R** together in one of the main scanning directions (rightward in the example of FIGS. **16A** to **16C**). As a result, the sheet bundle Pb is moved in the main scanning direction together with the internal tray **22**. The relative movement of the release claw **29** to a position away from the binding position **S3** allows the crimp binder **27** to crimp and bind the sheet bundle Pb.

(136) According to the third and fourth modifications, the binding position is crimped and bound by the crimp binder **27** even when the release claw **29** that does not move in the main scanning direction overlaps the binding position. As a result, a preferred binding strength is obtained for the

sheet bundle Pb. According to the third modification, since the side fences 26L and 26R are moved while the internal tray 22 is fixed, an assembly for moving the internal tray 22 may be omitted. On the other hand, according to the fourth modification, since the internal tray 22, the side fences 26L and 26R, and the sheet bundle Pb are moved together, the sheet bundle Pb is prevented from being bent on the internal tray 22.

(137) The embodiments of the present disclosure are applied to the crimp binder 27 that executes the edge binding as described above. However, the embodiments of the present disclosure may be applied to the saddle binder 33 that executes the saddle binding.

(138) The control method described above may be implemented by, for example, a program. In other words, the control method may be executed by a computer causing an arithmetic device, a storage device, an input device, an output device, and a control device to operate in cooperation with each other based on a program. The program may be written in, for example, a storage device or a storage medium and distributed. Alternatively, the program may be distributed through, for example, an electric communication line.

(139) Now, a description is given of some aspects of the present disclosure. According to a first aspect, a medium processing apparatus includes a conveyor, a receptacle, a conveyance-direction aligner, a crimp binder, and a controller. The conveyor conveys a sheet-shaped medium in a conveyance direction. The receptacle can hold a plurality of media, including the medium, conveyed by the conveyor. The conveyance-direction aligner contacts a downstream end, in the conveyance direction, of the plurality of media placed on the receptacle and aligns a position, in the conveyance direction, of the plurality of media. The crimp binder is disposed to face the downstream end, in the conveyance direction, of the plurality of media placed on the receptacle to press and deform a binding position on the plurality of media of which the position in the conveyance direction is aligned by the conveyance-direction aligner, to bind the plurality of media. The controller controls operations of the conveyor, the conveyance-direction aligner, and the crimp binder. The conveyance-direction aligner and the crimp binder are movable in a main scanning direction orthogonal to each of the conveyance direction and a thickness direction of the medium placed on the receptacle. The controller determines whether the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction, the controller moves the conveyance-direction aligner in the main scanning direction to a position away from the binding position. The controller then causes the crimp binder to crimp and bind the plurality of media.

(140) According to a second aspect, the medium processing apparatus of the first aspect further includes a pair of conveyance-direction aligners including the conveyance-direction aligner. The pair of conveyance-direction aligners is arranged opposite each other across a center position, in the main scanning direction, of the plurality of media. The controller increases or decreases an interval between the pair of conveyance-direction aligners while maintaining the pair of conveyance-direction aligners at equal distances from the center position.

(141) According to a third aspect, in the medium processing apparatus of the first or second aspect, the controller causes the crimp binder to sequentially crimp and bind a plurality of binding positions, including the binding position, arranged at given intervals in the main scanning direction.

(142) According to a fourth aspect, in the medium processing apparatus of the third aspect, the controller changes the intervals between the plurality of binding positions depending on a width of the plurality of media in the main scanning direction.

(143) According to a fifth aspect, in the medium processing apparatus of the third or fourth aspect, the controller causes the crimp binder to crimp and bind at least one binding position of the plurality of binding positions, the at least one binding position being away from the conveyance-direction aligner. The controller then moves the conveyance-direction aligner in the main scanning direction.

(144) According to a sixth aspect, in the medium processing apparatus of any one of the third to fifth aspects, the intervals between the plurality of binding positions are greater than 0 mm and less than 5 mm.

(145) According to a seventh aspect, the medium processing apparatus of any one of the first to sixth aspects further includes a releaser that contacts the downstream end, in the conveyance direction, of the plurality of media and moves upstream in the conveyance direction to release the plurality of media crimped and bound from the receptacle. The controller determines whether the releaser overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the releaser overlaps the binding position when the receptacle is viewed in the thickness direction, the controller relatively moves the releaser and the plurality of media to cause the releaser to be away from the binding position. The controller then causes the crimp binder to crimp and bind the binding position.

(146) According to an eighth aspect, the medium processing apparatus of the seventh aspect further includes a pair of width-direction aligners that contacts opposed ends, in the main scanning direction, of the plurality of media and moves in the main scanning direction to align a position, in the main scanning direction, of the plurality of media. The controller determines whether the releaser overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the releaser overlaps the binding position when the receptacle is viewed in the thickness direction, the controller moves the pair of width-direction aligners in the main scanning direction relative to the receptacle to move the plurality of media on the receptacle in the main scanning direction.

(147) According to a ninth aspect, the medium processing apparatus of the seventh aspect further includes a pair of width-direction aligners that contacts opposed ends, in the main scanning direction, of the plurality of media and moves in the main scanning direction to align a position, in the main scanning direction, of the plurality of media. The controller determines whether the releaser overlaps the binding position when the receptacle is viewed in the thickness direction. In a case where the controller determines that the releaser overlaps the binding position when the receptacle is viewed in the thickness direction, the controller moves the receptacle and the pair of width-direction aligners together in the main scanning direction to move the plurality of media together with the receptacle in the main scanning direction.

(148) According to a tenth aspect, in the medium processing apparatus of any one of the first to ninth aspects, the controller moves the conveyance-direction aligner in the main scanning direction to a position away from the binding position while the plurality of media is fixed to the receptacle.

(149) According to an eleventh aspect, the medium processing apparatus of the tenth aspect further includes a pair of width-direction aligners that contacts opposed ends, in the main scanning direction, of the plurality of media and moves in the main scanning direction to align a position, in the main scanning direction, of the plurality of media. The controller moves the conveyance-direction aligner in the main scanning direction while an interval between the pair of width-direction aligners is smaller than a width of the medium in the main scanning direction.

(150) According to a twelfth aspect, the medium processing apparatus of the tenth aspect further includes a contact and separation device that comes into contact with and separates from an uppermost surface of the plurality of media. The controller moves the conveyance-direction aligner in the main scanning direction while the contact and separation device is in contact with the uppermost surface of the plurality of media.

(151) According to a thirteenth aspect, the medium processing apparatus of any one of the first to twelfth aspects further includes a drive source and a driving force transmission assembly that moves the conveyance-direction aligner and the crimp binder in the main scanning direction by the driving source.

(152) According to a fourteenth aspect, the medium processing apparatus of any one of the first to twelfth aspects further includes a first driving source that moves the conveyance-direction aligner

in the main scanning direction and a second driving source that moves the crimp binder in the main scanning direction.

(153) According to a fifteenth aspect, an image forming system comprising an image forming apparatus and the medium processing apparatus of any one of the first to fourteenth aspects. The image forming apparatus forms an image on a medium. The medium processing apparatus crimps and binds the plurality of media on each of which the image is formed by the image forming apparatus.

(154) According to a sixteenth aspect, a medium processing method includes determining whether a conveyance-direction aligner overlaps a binding position when a receptacle is viewed in a thickness direction of a medium, moving the conveyance-direction aligner in a main scanning direction to a position away from the binding position based on a determination that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction of the medium, the main scanning direction being a direction orthogonal to each of a conveyance direction of the medium and the thickness direction of the medium placed on the receptacle, and moving crimp binder in the main scanning direction and causing the crimp binder to crimp and bind a plurality of media including the medium.

(155) According to a seventeenth aspect, a non-transitory recording medium stores a plurality of instructions which, when executed by one or more processors, causes the processors to perform a medium processing method. The method includes determining whether a conveyance-direction aligner overlaps a binding position when a receptacle is viewed in a thickness direction of a medium, moving the conveyance-direction aligner in a main scanning direction to a position away from the binding position based on a determination that the conveyance-direction aligner overlaps the binding position when the receptacle is viewed in the thickness direction of the medium, the main scanning direction being a direction orthogonal to each of a conveyance direction of the medium and the thickness direction of the medium placed on the receptacle, and moving crimp binder in the main scanning direction and causing the crimp binder to crimp and bind a plurality of media including the medium.

(156) According to one aspect of the present disclosure, selected positions in a main scanning direction on a bundle of media placed on a tray can be crimped and bound, and thus a preferred binding strength can be obtained.

(157) The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. It is therefore to be understood that the disclosure of this patent specification may be practiced otherwise by those skilled in the art than as specifically described herein and such modifications and alternatives are within the technical scope of the appended claims.

(158) Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

(159) The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose, processors, integrated circuits, application specific integrated circuits (ASICs), digital signal processors (DSPs), field programmable gate arrays (FPGAs), conventional circuitry and/or combinations thereof which are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein or otherwise known which is programmed or configured to carry out the recited functionality. When the hardware is a processor which may be considered a type of circuitry, the circuitry, means, or units

are a combination of hardware and software, the software being used to configure the hardware and/or processor.

Claims

1. A medium processing apparatus comprising: a conveyor including a conveyance roller configured to convey a sheet-shaped medium in a conveyance direction; a tray configured to hold the medium conveyed by the conveyor; a conveyance-direction fence configured to contact a downstream end, in the conveyance direction, of the medium placed on the tray and aligns a position, in the conveyance direction, of the medium; a crimp binder disposed to face the downstream end, in the conveyance direction, of the medium placed on the tray to press and deform a binding position on a plurality of media, including the medium, of which the position in the conveyance direction is aligned by the conveyance-direction fence, to bind the plurality of media; a pair of width-direction fences configured to contact opposed ends, in a main scanning direction, of the plurality of media and move in the main scanning direction to align a position, in the main scanning direction, of the plurality of media, the main scanning direction being orthogonal to each of the conveyance direction and a thickness direction of the medium placed on the tray; and circuitry configured to control operations of the conveyor, the conveyance-direction fence, and the crimp binder, the conveyance-direction fence and the crimp binder being movable in the main scanning direction, the circuitry being configured to: determine whether the conveyance-direction fence overlaps the binding position when the tray is viewed in the thickness direction; move the conveyance-direction fence in the main scanning direction to a position away from the binding position while the plurality of media is fixed to the tray, in a case where the circuitry determines that the conveyance-direction fence overlaps the binding position when the tray is viewed in the thickness direction; and cause the crimp binder to crimp and bind the plurality of media, and the circuitry being further configured to move the conveyance-direction fence in the main scanning direction while the opposed ends of the plurality of media contact the pair of width-direction fences.
2. The medium processing apparatus according to claim 1, further comprising a pair of conveyance-direction fences including the conveyance-direction fence, the pair of conveyance-direction fences arranged opposite each other across a center position, in the main scanning direction, of the plurality of media, wherein the circuitry is configured to increase or decrease an interval between the pair of conveyance-direction fences while maintaining the pair of conveyance-direction fences at equal distances from the center position.
3. The medium processing apparatus according to claim 1, wherein the circuitry is configured to cause the crimp binder to sequentially crimp and bind a plurality of binding positions, including the binding position, arranged at given intervals in the main scanning direction.
4. The medium processing apparatus according to claim 3, wherein the circuitry is configured to change the intervals between the plurality of binding positions depending on a width of the plurality of media in the main scanning direction.
5. The medium processing apparatus according to claim 3, wherein the circuitry is configured to cause the crimp binder to crimp and bind at least one binding position of the plurality of binding positions, the at least one binding position being away from the conveyance-direction fence, and move the conveyance-direction fence in the main scanning direction.
6. The medium processing apparatus according to claim 3, wherein the intervals between the plurality of binding positions are greater than 0 mm and less than 5 mm.
7. The medium processing apparatus according to claim 1, further comprising a contact configured to contact the downstream end, in the conveyance direction, of the plurality of media and move upstream in the conveyance direction to release the plurality of media crimped and bound from the tray, wherein the circuitry is configured to: determine whether the contact overlaps the binding

position when the tray is viewed in the thickness direction; relatively move the contact and the plurality of media to cause the contact to be away from the binding position in a case where the circuitry determines that the contact overlaps the binding position when the tray is viewed in the thickness direction; and cause the crimp binder to crimp and bind the binding position.

8. The medium processing apparatus according to claim 7, wherein the circuitry is configured to: determine whether the contact overlaps the binding position when the tray is viewed in the thickness direction; and move the pair of width-direction fences in the main scanning direction relative to the tray to move the plurality of media on the tray in the main scanning direction, in a case where the circuitry determines that the contact overlaps the binding position when the tray is viewed in the thickness direction.

9. The medium processing apparatus according to claim 7, wherein the circuitry is configured to: determine whether the contact overlaps the binding position when the tray is viewed in the thickness direction; and move the tray and the pair of width-direction fences together in the main scanning direction to move the plurality of media together with the tray in the main scanning direction, in a case where the circuitry determines that the contact overlaps the binding position when the tray is viewed in the thickness direction.

10. The medium processing apparatus according to claim 1, further comprising a contact and separation device including a tapping roller configured to come into contact with and separate from an uppermost surface of the plurality of media, wherein the circuitry is configured to move the conveyance-direction fence aligner in the main scanning direction while the contact and separation device is in contact with the uppermost surface of the plurality of media.

11. The medium processing apparatus according to claim 1, further comprising: a drive source; and a driving force transmission assembly including a pulley and a timing belt configured to move the conveyance-direction fence and the crimp binder in the main scanning direction by the driving source.

12. The medium processing apparatus according to claim 1, further comprising: a first driving source including a first motor configured to move the conveyance-direction fence in the main scanning direction; and a second driving source including a second motor configured to move the crimp binder in the main scanning direction.

13. An image forming system comprising: the medium processing apparatus according to claim 1, the medium processing apparatus being configured to crimp and bind the plurality of media on each of which an image is formed.
